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HACKATHON IDEA SUBMISSION

SovereignOps: Trust-Aware, Air-Gapped Incident Commander with Graph-Guided Remediation.

- **Team name:** Nexora
- **Team lead name:** Keerthan Marnad
- **USN of team members:** 1MS24AI029 , 1MS24AI030 ,1MS24AI060 , 1MS24AI054
- **Email ID and phone number of team lead:** keerthanmarnaddynun@gmail.com
(7204211541)
- **Track chosen:** Sovereign AI



What problem are you solving?

Incident triage speed determines downtime cost — but today's response workflows are unsafe, unverified, or both.

- **Problem Statement Explanation:** Enterprise incident response has 3 systemic failures — sensitive logs and topology can't be sent to cloud LLMs (sovereignty), local memory/RAG can't tell a verified fix from a hunch (fragile memory), and agents are either fully manual or dangerously autonomous (execution dilemma).
- **Who is affected:** IT and enterprise infrastructure teams — SREs and ops engineers who triage service crashes, network drops, and cascading database deadlocks under **zero-trust** and **compliance mandates**.
- **Why it matters:** Left unsolved: catastrophic downtime costs, compliance/data-sovereignty violations from cloud egress, and the real risk of an autonomous agent issuing a destructive command with no human check.



What is your solution?

SovereignOps — a 100% air-gapped, local-first engine that is your organization's Sovereign Second Brain.

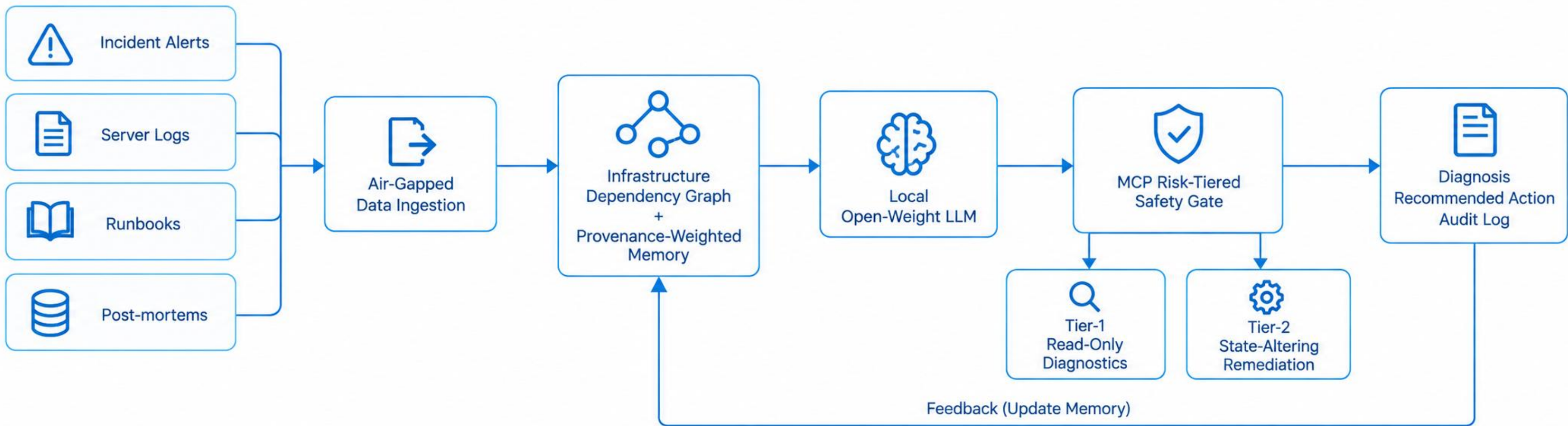
- **Solution description:** A local-first engine that diagnoses and remediates infrastructure incidents entirely on-premise — no telemetry, logs, or topology ever leave the organization's network.
- **How it works:** A 5-stage pipeline — Data (ingest alerts & runbooks), Knowledge (graph topology + embeddings), Memory (provenance-weighted store), Reasoning (local LLM traces root cause), Action (risk-tiered MCP execution) — with zero cloud egress at every stage.
- **Key features:** Graph-based root-cause traversal, provenance-tagged self-correcting memory, and a risk-tiered MCP safety gate — Tier-1 read-only diagnostics run autonomously, Tier-2 state-altering remediation requires human cryptographic sign-off.
- **Expected outcome:** Faster, verified triage; zero cloud data egress; and remediation that is auditable and never unconstrained.



System Architecture

SovereignOps — System Architecture

Local AI for Incident Memory & Response





What makes your idea different?

It replaces naive search and blind autonomy with graph reasoning and calibrated trust.

- **What is different:** Graph-augmented root cause analysis instead of naive semantic search, provenance-weighted memory that self-corrects when past retrospectives are updated, and deterministic risk-tiered execution boundaries.
- **Current alternative:** Teams either paste sensitive logs into a cloud LLM copilot, or use append-only local RAG stores that treat every past fix as equally trustworthy, or run agents that are purely advisory (manual copy-paste) or fully autonomous with no guardrails.
- **Why it matters:** Organizations get enterprise-ready incident response that is both fast and safe — sovereign by design, self-correcting over time, and incapable of an unsupervised destructive action.



Can you build it?

A working air-gapped triage loop, demoed end-to-end on a simulated incident.

- **Tech stack:** Local open-weight LLM (Ollama / llama.cpp / vLLM); Graphiti for dependency-graph traversal using BFS/DFS cascade traversal, Dijkstra shortest-path fault localization, and PageRank-weighted node criticality; PostgreSQL/pgvector for embeddings; Zep/Mem0/Letta-style provenance memory; and MCP for tiered tool execution.
- **Core memory indexing:** Core provenance-memory indexing and lookup algorithms are written in C to optimize time complexity at scale.
- **MVP scope:** Ingest a syslog/alert → traverse a sample dependency graph → retrieve provenance-tagged runbooks → run Tier-1 read-only diagnostics automatically → surface a Tier-2 remediation for human sign-off.
- **Demo-day deliverable:** A live, fully local incident walkthrough — alert in, graph-traced root cause out, autonomous diagnostics executed, and a remediation action awaiting a cryptographic human sign-off.



What will you achieve?

Faster, auditable incident response — without ever trusting a third party with your infrastructure.

- **Expected impact:** IT/SRE teams cut mean-time-to-resolution on outages while staying fully compliant with zero-trust and data-ownership mandates — no logs or topology ever leave the org.
- **Success metric:** Reduction in root-cause diagnosis time, % of Tier-1 diagnostics resolved autonomously, and accuracy of graph-traced root cause vs. actual cause.
- **Demo outcome:** An incident diagnosed end-to-end on local infrastructure — root cause traced through the dependency graph, a verified fix retrieved from memory, and a remediation action gated on human sign-off, with zero data ever leaving the machine.